

Claims Denial Risk: Model & GenAI Explanations

Problem

The review team can inspect only the top 25% of claims by risk score before submission -- everything past that cutoff ships without a human look. That constraint, not accuracy, drives the metric choice: a false negative (a claim that will be denied but scores low enough to skip review) costs a full denial-and-resubmit cycle, while a false positive (a claim that scores high but would have been paid) only costs a few minutes of an analyst's attention. Since review capacity is fixed at 25% regardless of threshold, the headline metric is recall at the top 25% of claims ranked by score -- effectively, the fraction of actual denials the review team would catch given their fixed capacity. This is reported alongside ROC-AUC and PR-AUC as secondary sanity checks, but model selection itself was driven by validation-set recall@25%, not by accuracy or AUC.

Data Findings

Full univariate and bivariate analysis, with an inference note after every chart, is in [notebooks/eda.ipynb](#).

High-Leverage Manual Checks

Two specific features flag near-guaranteed denials: claims missing required prior authorization (46% denial rate vs. 22% baseline) and claims missing supporting documentation (39% denial rate). These are checkable before submission and represent the highest-leverage manual checks available.

Payer Variability

Denial rates vary substantially by payer, ranging from 11% to 37% across the 12 payers in the training history. This individual payer behavior is a significant signal, justifying the inclusion of *payer_id* as a one-hot encoded model feature.

Denial Reasons & Distribution

Causes are widely distributed: Payer policy (20%), missing documentation (17%), eligibility (13%), late filing (12%), missing authorization (11%), coding/network errors (10% each), and missing referrals (7%). The diversity of causes suggests the review team requires a comprehensive checklist rather than a single solution.

Numeric Predictors & Correlation

Numeric fields like total billed amount and code counts are weak predictors (correlation < 0.13). The primary exception is *days_to_submit*, where denied claims average 22 days versus 19 for paid claims. Additionally, billed and expected payments are nearly collinear ($r=0.95$), a redundancy handled by the engineered *payment_ratio* feature.

Models

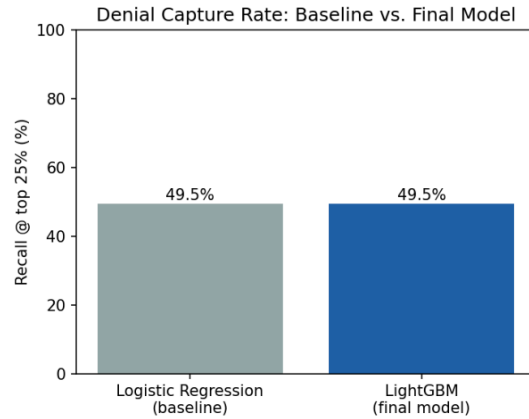
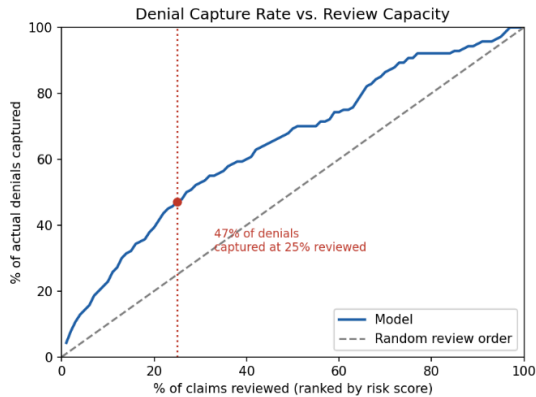
Two models were trained on the train split: a class-weighted Logistic Regression baseline, and a LightGBM classifier.

LightGBM's hyperparameters (*max_depth*, *num_leaves*, *min_child_samples*, *n_estimators*) were grid searched over 16 configurations, all selected on validation *recall@25%* -- the same metric used for the final baseline-vs-LightGBM choice, so the tuning objective matches what the model is actually picked for. An earlier, unregularized default (deeper trees, no *min_child_samples* floor) scored below the linear baseline -- a sign of overfitting given the ~2,100-row training set -- which is why the grid is restricted to shallow configs (*max_depth*≤3, *num_leaves*≤8); *recall@25%* across the grid ranges from 0.427 to 0.495 (best: *max_depth*=2, *num_leaves*=4, *min_child_samples*=50, *n_estimators*=100; full sweep in *outputs/lightgbm_grid_search.csv*). The tuned LightGBM matched but did not beat the Logistic Regression baseline.

Logistic Regression was kept as the final model for its interpretability and stability at this dataset size; the comparison is a genuine finding, not a fallback -- more data or feature engineering would likely be needed before a gradient-boosted model earns its added complexity here. High/Medium/Low risk tiers use fixed probability cutoffs taken from the validation score distribution's 75th and 50th percentiles (0.594 and 0.418), then applied unchanged to test and to *current_claims.csv*. A fixed operating point keeps the 'top 25%' definition stable across batches instead of silently redefining it based on whatever claims happen to be scored that day.

Test Results

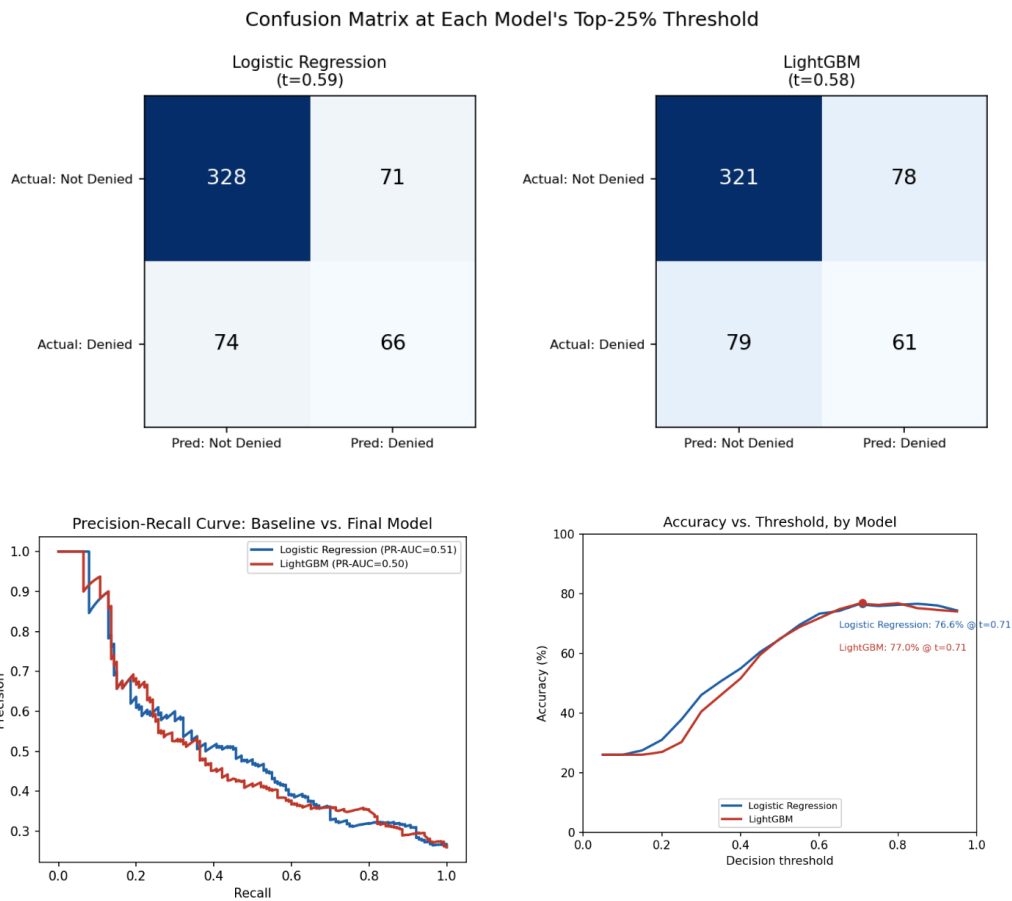
On the held-out test split (539 claims, 140 actual denials): ROC-AUC 0.70, PR-AUC 0.51. At the top 25% cutoff, the model captures 47% of all actual denials while reviewing only 25% of claims -- **roughly double the ~25%** a random review order would catch. Precision within that top quartile is 49%, meaning about half of the claims flagged for review are in fact denied.



Left: denial capture rate vs. random review order, with the 25%-reviewed operating point marked.

Right: validation recall@25% baseline vs. final model -- the near-identical bars are why Logistic Regression, not LightGBM, was kept as the final model.

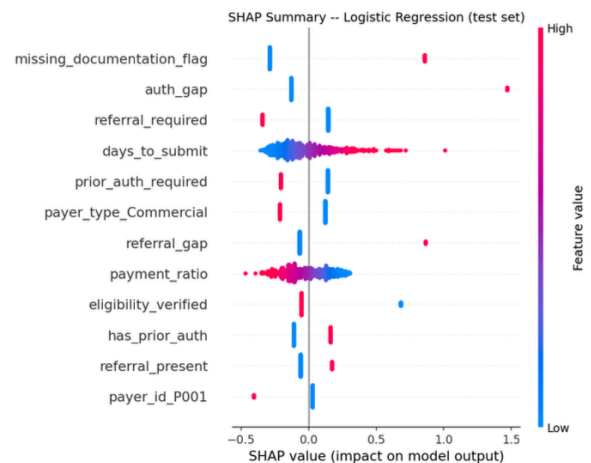
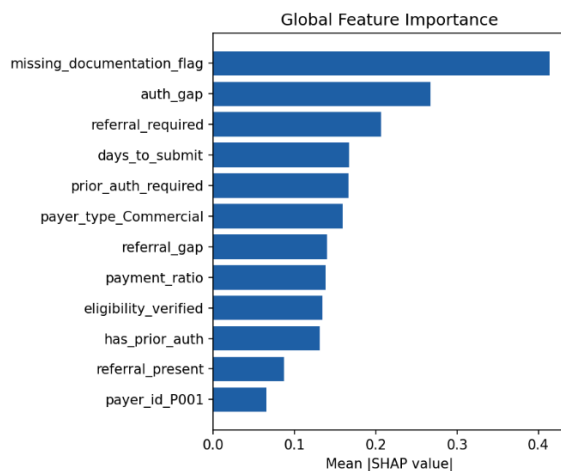
Confusion Matrix, PR-AUC, and Threshold/Accuracy -- Both Models



Threshold-vs-accuracy sweep (full table in outputs/threshold_sweep.csv): accuracy for both models peaks around $t=0.71$, at 76.6% (Logistic Regression) and 77.0% (LightGBM) -- notably higher than the ~73% accuracy at the operational top-25% threshold. This is expected, not a problem: accuracy is maximized by a threshold that predicts 'not denied' for most claims (the majority class), which would flag far fewer than 25% of claims for review and defeats the point of a review-priority score. Accuracy is reported here for completeness but was never the selection criterion (Section 1).

Feature Importance and SHAP Analysis

- Global feature importance, based on mean |SHAP value| across the test set, confirms that *missing_documentation_flag* and *auth_gap* are the primary drivers of predicted risk.
- Secondary drivers include *referral_required*, *days_to_submit*, and *prior_auth_required*.
- The SHAP summary plot reveals that high feature values for documentation and authorization gaps sitting almost entirely on the positive-SHAP side, reliably pushing risk upward.
- This computation populates the per-claim *top_risk_factors* and grounds the LLM explanations.



5. LLM Explanations

For the 10 highest-risk current claims, an LLM (OpenAI gpt-4o-mini) generates analyst-facing explanations based on the following process:

- **Grounding:** Explanations are derived strictly from the claim's field values and SHAP-derived risk factors to prevent hallucinations.
- **Constraints:** The output is restricted to 2-3 sentences in plain language with exactly one concrete recommended action.
- **Disclaimer:** Each note must explicitly state that the score is a risk estimate, not a certainty.

Prompt Template:

SYSTEM: You are writing short risk notes for a hospital billing analyst. Rules: use ONLY the facts given below; plain English, no jargon; exactly one concrete recommended action; state this is a risk estimate, not a guarantee; 2-3 sentences, no preamble/headers/bullets.

USER: Claim ID: {claim_id} | Payer type: {payer_type} | Visit type: {visit_type} | Total billed: \${total_billed} | Model's predicted denial risk: {denial_probability} ({risk_tier}) | Top risk factors: {top_risk_factors}

Example Output (Highest-Risk):

This claim is flagged as high risk (95% predicted denial probability), mainly because a referral is required but not on file and required supporting documentation appears to be missing. Recommended action: verify a referral exists and attach it before submission. This is a model-based risk estimate, not a guarantee the claim will be denied.

Low-Risk Sanity Check:

This claim is flagged as low risk (10% predicted denial probability); the model did not find a specific documentation, authorization, or eligibility gap driving this score. Recommended action: no special pre-submission review needed -- proceed with normal submission. This is a model-based risk estimate, not a guarantee the claim will be paid.

6. Honest Limitation

Due to active quota limits on the OpenAI API key, these explanations were produced by a manual-drafting fallback in *explain.py* rather than a live LLM call. This fallback uses the same risk-factor data but lacks the fluency and nuance of a live model. Swapping to a funded API key requires no code changes as the system attempts the API call first before falling back.

7. What I'd Improve With More Time

- Add a per-payer confidence flag for lower-volume payers to account for sparser data.
- Implement an automated groundedness check to verify all LLM facts against the input data.
- Calibrate probabilities using Platt scaling or isotonic regression to improve estimation accuracy.
- Revisit gradient boosting models (LightGBM/XGBoost) as historical dataset size increases.

